

7. (a) **Complete the following table** about the processes by which substances move through cell membranes. [3]

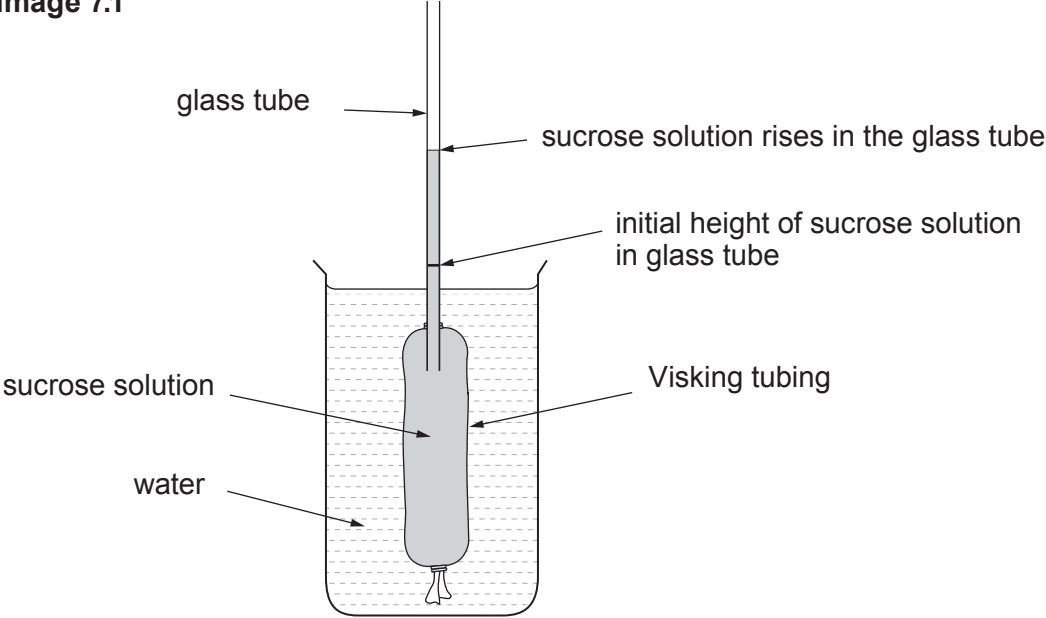
Place a tick (✓) or a cross (✗) in each box to indicate if the statement applies to each process or not.

	Active transport	Osmosis	Diffusion
Energy (ATP) needed			
Against a concentration gradient			
Down a concentration gradient			



- (b) The apparatus shown in **Image 7.1** was set up using a piece of Visking tubing filled with sucrose solution. The Visking tubing was knotted at its bottom end and tied at its top end to a length of glass tube. The Visking tubing was then placed in a beaker of water and left for 1 hour. At the end of this time the sucrose solution had risen up the tube.

Image 7.1



Explain why the sucrose solution moved up the glass tube.

[4]

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